

Black Box Testing - DentWise Application

1. Testing Approach

This document outlines the black-box testing strategy using: - **Equivalence Partitioning (EP)**: Dividing input data into valid and invalid partitions - **Boundary Value Analysis (BVA)**: Testing at the edges of input ranges

2. Test Cases for formatPhoneNumber Function

2.1 Equivalence Partitioning Analysis

Partition	Description	Valid/Invalid	Example Input
EP1	Empty string	Valid	""
EP2	1-3 digits	Valid	"011"
EP3	4-6 digits	Valid	"011245"
EP4	7-9 digits	Valid	"011245552"
EP5	10-11 digits	Valid	"01124555246"
EP6	More than 11 digits	Valid (truncated)	"011245552461234"
EP7	Contains letters	Valid (stripped)	"011abc245"
EP8	Contains special chars	Valid (stripped)	"(+2) 011-245"

2.2 Boundary Value Analysis

Boundary	Description	Input	Expected Output
BVA1	Min length (0)	""	""
BVA2	Min+1 (1 digit)	"0"	"(+2) 0"
BVA3	Partition edge (3 digits)	"011"	"(+2) 011"
BVA4	Partition edge (4 digits)	"0112"	"(+2) 011 2"
BVA5	Partition edge (6 digits)	"011245"	"(+2) 011 245"
BVA6	Partition edge (7 digits)	"0112455"	"(+2) 011 245 5"
BVA7	Partition edge (9 digits)	"011245552"	"(+2) 011 245 552"
BVA8	Partition edge (10 digits)	"0112455524"	"(+2) 011 245 552 4"
BVA9	Max valid (11 digits)	"01124555246"	"(+2) 011 245 552 46"
BVA10	Max+1 (12 digits)	"011245552461"	"(+2) 011 245 552 46"

3. Detailed Test Case Table

TC-BB-01: Empty String Input

Field	Value
Test Case ID	TC-BB-01
Description	Verify empty string returns empty string
Preconditions	None
Test Data	Input: ""

Field	Value
Steps	1. Call formatPhoneNumber("")
Expected Result	Returns ""
Actual Result	""
Status	☐ PASS
Technique	EP1, BVA1

TC-BB-02: Single Digit Input

Field	Value
Test Case ID	TC-BB-02
Description	Verify single digit is formatted correctly
Preconditions	None
Test Data	Input: "0"
Steps	1. Call formatPhoneNumber("0")
Expected Result	Returns "(+2) 0"
Actual Result	"(+2) 0"
Status	☐ PASS
Technique	BVA2

TC-BB-03: Three Digit Input

Field	Value
Test Case ID	TC-BB-03
Description	Verify 3-digit number is formatted with country code
Preconditions	None
Test Data	Input: "011"
Steps	1. Call formatPhoneNumber("011")
Expected Result	Returns "(+2) 011"
Actual Result	"(+2) 011"
Status	☐ PASS
Technique	EP2, BVA3

TC-BB-04: Six Digit Input

Field	Value
Test Case ID	TC-BB-04
Description	Verify 6-digit number is formatted with proper spacing
Preconditions	None
Test Data	Input: "011245"
Steps	1. Call formatPhoneNumber("011245")
Expected Result	Returns "(+2) 011 245"
Actual Result	"(+2) 011 245"
Status	☐ PASS
Technique	EP3, BVA5

TC-BB-05: Nine Digit Input

Field	Value
Test Case ID	TC-BB-05
Description	Verify 9-digit number is formatted correctly
Preconditions	None
Test Data	Input: "011245552"
Steps	1. Call formatPhoneNumber("011245552")
Expected Result	Returns "(+2) 011 245 552"
Actual Result	"(+2) 011 245 552"
Status	☐ PASS
Technique	EP4, BVA7

TC-BB-06: Full Egyptian Phone Number (11 digits)

Field	Value
Test Case ID	TC-BB-06
Description	Verify complete Egyptian phone number formatting
Preconditions	None
Test Data	Input: "01124555246"
Steps	1. Call formatPhoneNumber("01124555246")
Expected Result	Returns "(+2) 011 245 552 46"
Actual Result	"(+2) 011 245 552 46"
Status	☐ PASS
Technique	EP5, BVA9

TC-BB-07: Input with Special Characters

Field	Value
Test Case ID	TC-BB-07
Description	Verify special characters are stripped
Preconditions	None
Test Data	Input: "(+2) 011-245-5524"
Steps	1. Call formatPhoneNumber("(+2) 011-245-5524")
Expected Result	Returns "(+2) 011 245 552 4"
Actual Result	"(+2) 011 245 552 4"
Status	☐ PASS
Technique	EP8

TC-BB-08: Input with Spaces

Field	Value
Test Case ID	TC-BB-08
Description	Verify spaces in input are handled
Preconditions	None
Test Data	Input: "011 245 552"
Steps	1. Call formatPhoneNumber("011 245 552")

Field	Value
Expected Result	Returns "(+2) 011 245 552"
Actual Result	"(+2) 011 245 552"
Status	☐ PASS
Technique	EP8

TC-BB-09: Four Digit Input (Boundary)

Field	Value
Test Case ID	TC-BB-09
Description	Verify 4-digit boundary is handled
Preconditions	None
Test Data	Input: "0112"
Steps	1. Call formatPhoneNumber("0112")
Expected Result	Returns "(+2) 011 2"
Actual Result	"(+2) 011 2"
Status	☐ PASS
Technique	BVA4

TC-BB-10: Seven Digit Input (Boundary)

Field	Value
Test Case ID	TC-BB-10
Description	Verify 7-digit boundary is handled
Preconditions	None
Test Data	Input: "0112455"
Steps	1. Call formatPhoneNumber("0112455")
Expected Result	Returns "(+2) 011 245 5"
Actual Result	"(+2) 011 245 5"
Status	☐ PASS
Technique	BVA6

4. Test Execution Summary

Total Test Cases	Passed	Failed	Pass Rate
10	10	0	100%

5. Additional Test Cases (Other Functions)

getNext5Days Function

TC ID	Description	Expected	Status
TC-DATE-01	Returns array of 5 dates	Array.length = 5	☐ PASS
TC-DATE-02	Dates in ISO format	YYYY-MM-DD pattern	☐ PASS
TC-DATE-03	First date is tomorrow	Today + 1	☐ PASS

getAvailableTimeSlots Function

TC ID	Description	Expected	Status
TC-SLOT-01	Returns 12 slots	Array.length = 12	☐ PASS
TC-SLOT-02	First slot is 09:00	"09:00"	☐ PASS
TC-SLOT-03	Last slot is 16:30	"16:30"	☐ PASS
TC-SLOT-04	HH:MM format	Pattern match	☐ PASS

generateAvatar Function

TC ID	Description	Expected	Status
TC-AVATAR-01	Male avatar URL	Contains "boy"	☐ PASS
TC-AVATAR-02	Female avatar URL	Contains "girl"	☐ PASS
TC-AVATAR-03	Multiple spaces handled	Spaces removed	☐ PASS

6. Jira Test Case Upload Format

Test Case ID: TC-BB-XX
 Summary: [Brief description]
 Priority: Medium
 Test Type: Functional
 Preconditions: [Any setup needed]
 Test Steps:
 1. [Step description]
 Expected Result: [What should happen]
 Actual Result: [What happened]
 Status: Pass/Fail
 Attachments: [Screenshots if needed]